```python
# DAY 1 - Variables, Data Types, I/O, Operator

# 1. Sum of Two Numbers

print("1. Sum of Two Numbers")
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Sum =", num1 + num2)

print("\n------------------")



# 2. Area of a Rectangle

print("2. Area of Rectangle")
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))
area = length * breadth
print("Area =", area)

print("\n------------------")


# 3. Simple Interest

print("3. Simple Interest")
p = float(input("Enter Principal Amount: "))
r = float(input("Enter Rate of Interest: "))
t = float(input("Enter Time (years): "))
si = (p * r * t) / 100
print("Simple Interest =", si)

print("\n------------------")
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# 4. Celsius to Fahrenheit

print("4. Celsius to Fahrenheit")
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32
print("Temperature in Fahrenheit =", fahrenheit)

print("\n-------------------")



# 5. Employee Salary Calculation

print("5. Employee Salary Calculation")
basic_salary = float(input("Enter Basic Salary: "))

hra = basic_salary * 20 / 100
da = basic_salary * 10 / 100
gross_salary = basic_salary + hra + da

print("HRA =", hra)
print("DA =", da)
print("Gross Salary =", gross_salary)
```

```
1. Sum of Two Numbers
Enter first number: 1
Enter second number: 3
Sum = 4.0

-------------------
2. Area of Rectangle
Enter length: 2.3
Enter breadth: 4.3
Area = 9.889999999999999

-------------------
3. Simple Interest
Enter Principal Amount: 20000
Enter Rate of Interest: 3
Enter Time (years): 5
Simple Interest = 3000.0

-------------------
4. Celsius to Fahrenheit
Enter temperature in Celsius: 34
Temperature in Fahrenheit = 93.2

-------------------
5. Employee Salary Calculation
Enter Basic Salary: 30000
HRA = 6000.0
DA = 3000.0
Gross Salary = 39000.0
```

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# DAY 2 - CONDITIONS


# 6. Check Positive, Negative, or Zero

num = int(input("Enter a number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")

print("="*50)

# 7. Largest Among Two Numbers

a = int(input("\nEnter first number: "))
b = int(input("Enter second number: "))
if a > b:
    print("Largest =", a)
else:
    print("Largest =", b)

print("="*50)
```

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# 8. Largest Among Three Numbers

a = int(input("\nEnter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
    print("Largest =", a)
elif b >= a and b >= c:
    print("Largest =", b)
else:
    print("Largest =", c)

print("="*50)

# 9. Check Even or Odd

num = int(input("\nEnter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")

print("="*50)

# 10. Check Leap Year

year = int(input("\nEnter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

```
Enter a number: 3
Positive
==================================================

Enter first number: 5
Enter second number: 7
Largest = 7
==================================================

Enter first number: 3
Enter second number: 5
Enter third number: 4
Largest = 5
==================================================

Enter a number: 2
Even
==================================================

Enter a year: 9
Not a Leap Year
```

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# DAY 3 - LOOPS

# 11. Print Numbers from 1 to 100
print("\nNumbers from 1 to 100:")
for i in range(1, 101):
    print(i, end=" ")

print("="*50)

# 12. Print Even Numbers Between 1 and 100

print("\n\nEven Numbers from 1 to 100:")
for i in range(2, 101, 2):
    print(i, end=" ")

print("="*50)

# 13. Sum of First N Natural Numbers

n = int(input("\n\nEnter N: "))
sum_n = 0
for i in range(1, n + 1):
    sum_n += i
print("Sum =", sum_n)

print("="*50)
```

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# 14. Multiplication Table

num = int(input("\nEnter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")

print("="*50)

# 15. Factorial of a Number

num = int(input("\nEnter a number: "))
fact = 1
for i in range(1, num + 1):
    fact *= i
print("Factorial =", fact)
```

```
Numbers from 1 to 100:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 5
6 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 ======================
============================


Even Numbers from 1 to 100:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100 ==========
========================================


Enter N: 3
Sum = 6
==================================================

Enter a number: 4
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
==================================================

Enter a number: 6
Factorial = 720
```

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# DAY 4 - FUNCTIONS

# 16. Function to find square of a number

def square(n):
    return n * n

num = int(input("Enter a number: "))
print("Square =", square(num))

print("="*50)

# 17. Function to find maximum of two numbers

def maximum(a, b):
    return a if a > b else b

a = int(input("\nEnter first number: "))
b = int(input("Enter second number: "))
print("Maximum =", maximum(a, b))

print("="*50)

# 18. Function to calculate simple interest

def simple_interest(p, r, t):
    return (p * r * t) / 100

p = float(input("\nEnter Principal: "))
r = float(input("Enter Rate: "))
t = float(input("Enter Time: "))
print("Simple Interest =", simple_interest(p, r, t))

print("="*50)
```

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# 19. Function to check even or odd

def check_even_odd(n):
    if n % 2 == 0:
        return "Even"
    return "Odd"

num = int(input("\nEnter a number: "))
print(check_even_odd(num))

print("="*50)

# 20. Function to calculate area of a circle

def area_circle(radius):
    return 3.14 * radius * radius

radius = float(input("\nEnter radius: "))
print("Area of Circle =", area_circle(radius))
```

```
Enter a number: 1
Square = 1
==================================================

Enter first number: 4
Enter second number: 3
Maximum = 4
==================================================

Enter Principal: 50000
Enter Rate: 4
Enter Time: 3
Simple Interest = 6000.0
==================================================

Enter a number: 5
Odd
==================================================

Enter radius: 4
Area of Circle = 50.24
```

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# DAY 5 - LISTS, TUPLES, DICTIONARY, SET

# 21. Create a list of 10 numbers and print all elements

numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
print("\nList Elements:")
for num in numbers:
    print(num, end=" ")


# 22. Find largest element in a list

print("\n\nLargest Element =", max(numbers))

print("="*50)

# 23. Calculate sum of all elements in a list

print("Sum of Elements =", sum(numbers))

print("="*50)

# 24. Count even numbers in a list

count = 0
for num in numbers:
    if num % 2 == 0:
        count += 1

print("Count of Even Numbers =", count)

print("="*50)
```

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# 25. Program to remove duplicate elements using sets

numbers = [10, 20, 30, 20, 40, 10, 50, 30]

unique_numbers = list(set(numbers))

print("Original List:", numbers)
print("List after removing duplicates:", unique_numbers)
```

```
List Elements:
1 2 3 4 5 6 7 8 9 10

Largest Element = 10
==================================================
Sum of Elements = 55
==================================================
Count of Even Numbers = 5
==================================================
Original List: [10, 20, 30, 20, 40, 10, 50, 30]
List after removing duplicates: [40, 10, 50, 20, 30]
```

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# 26. Reverse a Number

num = int(input("Enter a number: "))
rev = int(str(num)[::-1])
print("Reversed Number =", rev)

print("="*50)


# 27. Check Palindrome Number

num = input("Enter a number: ")
if num == num[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")

print("="*50)


# 28. Count Digits in a Number

num = input("Enter a number: ")
print("Number of digits =", len(num))

print("="*50)

# 29. Sum of Digits
num = input("Enter a number: ")
total = 0
for i in num:
    total += int(i)
print("Sum of digits =", total)

print("="*50)
```

```python
# 30. Fibonacci Series

n = int(input("Enter number of terms: "))
a, b = 0, 1
print("Fibonacci Series:")
for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
print()

print("="*50)


# 31. Check Prime Number

num = int(input("Enter a number: "))
prime = True

if num <= 1:
    prime = False
else:
    for i in range(2, num):
        if num % i == 0:
            prime = False
            break

if prime:
    print("Prime Number")
else:
    print("Not a Prime Number")

print("="*50)
```

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# 32. Prime Numbers Between 1 and 100

print("Prime Numbers from 1 to 100:")
for num in range(2, 101):
    prime = True
    for i in range(2, num):
        if num % i == 0:
            prime = False
            break
    if prime:
        print(num, end=" ")
print()

print("="*50)


# 33. Count Vowels in a String

text = input("Enter a string: ")
count = 0

for ch in text.lower():
    if ch in "aeiou":
        count += 1

print("Number of vowels =", count)

print("="*50)
```

```python
# 34. Reverse a String

text = input("Enter a string: ")
print("Reversed String =", text[::-1])

print("="*50)


# 35. Count Frequency of Each Character

text = input("Enter a string: ")

for ch in set(text):
    print(ch, "=", text.count(ch))
```

```
Enter a number: 4
Reversed Number = 4
==================================================
Enter a number: 3
Palindrome
==================================================
Enter a number: 2
Number of digits = 1
==================================================
Enter a number: 5
Sum of digits = 5
==================================================
Enter number of terms: 4
Fibonacci Series:
0 1 1 2
==================================================
Enter a number: 3
Prime Number
==================================================
Prime Numbers from 1 to 100:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
==================================================
Enter a string: 5
Number of vowels = 0
==================================================
Enter a string: 4
Reversed String = 4
==================================================
Enter a string: 3
3 = 1
```